

Morteza Shoushtari

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PROFESSIONAL SUMMARY

Senior Network/System Engineer specializing in AI/HPC network infrastructure at Supermicro, backed by a Ph.D. in Computer and Electrical Engineering. Deep expertise across wire networking (DC and Enterprise), wireless, and cellular network. Passionate about working with great teams and always learning in fast-moving technical environments.

SKILLS

- Networking & Protocols

Wired/Wireless/Data Center Network Architecture & Design | TCP/IP, IPv4/IPv6, FLSM/VLSM | L2 Switching (VLAN, STP, LACP) | L3 Routing (EIGRP, OSPF, BGP, MPLS) | 802.11 (WiFi), 802.15 (Bluetooth, Zigbee) | Cellular Networks (LTE/5G) | Protocol Stack (PHY, MAC, RLC, PDCP, RRC, SDAP)

- AI/HPC Infrastructure

AI/HPC Network Design & Validation | InfiniBand/Ethernet Networking | RDMA, RoCE | Fiber Optics & Transceivers | High-Speed Interconnects & Cabling Architecture

- Security & Coding

Cryptographic Algorithms (AES, DES, RSA) | Post-Quantum Cryptosystems | Physical Layer Security | Error-Correction Coding (LDPC, Polar, Turbo, Viterbi)

- Systems & Tools

Server Operating Systems (Mac, Windows, Linux) | Virtualization Technologies | Network Troubleshooting & Analysis (Wireshark) | Process Automation

- Programming Languages

C++, MATLAB, Python

- Certification

Supermicro GPU Service Associate/Specialist/Expert (SGSA, SGSS, SGSE), Supermicro Direct Liquid Cooling Specialist, NVIDIA InfiniBand, CCNA, HCNA, MCITP, Network+, A+

WORK EXPERIENCE

Super Micro Computer, Inc.

CA, USA

Sr. Network/System Engineer

May 2024 – Present

- Validated network infrastructure for hyperscale AI/HPC deployments, including xAI's Colossus cluster (200k+ NVIDIA H100/H200 GPUs), a 4k-GPU H200 deployment for GBM in Dubai, and a 20k-GPU GB300 deployment for Mistral in Paris, spanning InfiniBand and Ethernet fabrics.
- Reduced deployment errors by 60% by developing and implementing rigorous port-mapping and network-compatibility validation procedures across multi-vendor cabling and transceiver configurations.
- Engineered port connectivity and shuffle solutions (cables/modules) and high-speed interconnect architectures to ensure signal integrity and scalability across racks of varying GPU generations (B200, GB300, VR and AMD MI350/MI450) and interconnect standards.
- Lead network and system validation activities, developing test plans, validation procedures, and technical documentation for network rack components.
- Collaborated with cable, transceiver, and switch vendors to diagnose and resolve complex interoperability issues, minimizing downtime across mission-critical AI infrastructure rollouts.
- Collaborated cross-functionally with datacenter architects, ODM/OEM partners, and deployment teams across the US, Europe and Asia to align on product requirements and ensure successful large-scale rollouts.

Brigham Young University

Utah, USA

Graduate Research Assistant

Jan 2019 – July 2023

- Discovered new properties of secrecy coding from the perspectives of information and coding theory.
- Devised a high-speed algorithm to construct the optimal code in Nested Linear Secrecy Codes, enhancing code identification and selection efficiency.

- Conducted a detailed assessment and visualized secrecy problems such as eavesdropping in aeronautical mobile telemetry communication and suggested the use of secrecy coding and specialized version of post-quantum cryptosystems for this type of communication systems.

HUAWEI

Network Engineer

Tehran, Iran

Aug 2016 – Aug 2017

- Designed and implemented wired/wireless networks, and performed network maintenance and system upgrades.
- Maximized network performance and increased network availability by 99.99% by implementing a redundant network architecture.
- Performed installations and configurations, and support of various network devices such as switches (Huawei, Cisco), wireless access points (Huawei, Aruba), cameras, and video conferencing systems.

JYAN

Network Engineer

Tehran, Iran

Jan 2016 – Aug 2016

- Managed wired communication networks, and performed network maintenance and system upgrades.
- Monitor performance and ensure system availability and reliability.
- Performed installations and configurations of various network devices such as switches (Cisco), wireless access points (Aruba), cameras, and video conferencing systems.

Persia Cloud

Network Engineer

Tehran, Iran

Jun 2015 – Jan 2016

- Performed installations and configurations of HP servers, virtual machines (VMs), and Cisco switches.
- Administered SaaS and PaaS layers of the company's cloud platforms (Citrix, IIRAS).
- Managed and configured Microsoft application servers CRM, SharePoint, and Lync.

EDUCATION

Brigham Young University

Ph.D. in Computer and Electrical Engineering

UT, USA

Jan 2019 – July 2023

Shiraz University

M.Sc. in Information Technology

Fars, Iran

2011 – 2013

SELECTED PUBLICATION

- “On Best Erasure Wiretap Codes: Equivocation Matrices and Design Principles”, Entropy Journal, 2025.
- “Secrecy Coding for the Binary Symmetric Wiretap Channel via Linear Programming”, IEEE Transactions on Information Forensics and Security, 2025.
- “Optimizing Finite Blocklength Nested Linear Codes: Using the Worst Code to Find the Best Code”, Entropy Journal, 2023.
- Full list available on request

AWARDS & ACHIEVEMENTS

- Best paper award, International Telemetry Conference, 2022.
- Second best graduate student paper award, International Telemetry Conference, 2021.
- Third place award, IEEE Intermountain Engineering, Technology, and Computing Conference (IEEE i-ETC), 2023.
- Outstanding IT engineer in Huawei Technologies Company, 2017.

REFERENCES

Available upon request.